

Lab Tasks

Composition & Aggregation

1. Make a calculator class that has its own display class as a member object, use that display class for displaying calculation results and to display last result (composition),
2. Make a calculator class that is shared by multiple students (aggregation)

Runtime polymorphism

Create a base class Shape with virtual double area and override this in Rectangle and Circle class to compute actual values, use shape* pointer to access diff shapes

Compile time Polymorphism

1. Make a calculator class with overloaded add (), multiply(), subtract(), divide() functions (function overloading)
2. Create a Distance class with feet and inches as members, overload the == operator and display whether two distances are equal or not (operator overloading)

Friend Functions & PVFs

1. Make Rectangle and Circle class that have a friend function to calculate their combined totalarea() (friend function)
2. A Shape class that contains a pure virtual function (=0) draw(), override that function in Circle and Rectangle class with their own implementation of this function(pure virtual functions)

Static Data Members and Functions

1. Make a class that increments static member count whenever a new object is made, use it to demonstrate how a static member is shared across all objects of the same class (static data member)
2. Change the first program by making a static print() function to print the total number of objects made so far (static function)

File Handling

1. Create a file named notes.txt, write lines to it, open it read mode and display all file content then append the file by adding name and roll number
2. Count number of lines in an existing file
3. Copy one file into another file
4. Write student details such as name and roll number of 3 students to a file and then read them to display content on screen

File Pointers

All 4 examples of file pointers from lecture no. 10

Exception Handling & Templates

1. A program to handle dividing by zero exception
2. A program where you take an array of size 3 and initialize it equal to 0, ask the user for an index and the value he wants to store at that index, throw error if the index is less than 0 or greater than or equal to three,
3. A calculator template (also use exception handling where required)
4. A logger class using template